

Three fixed points on the symplectic quadric threefold

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Abstract

We construct a smooth Hamiltonian diffeomorphism of the complex quadric threefold with exactly three fixed points, at least one of which is degenerate. The critical number and the unit-inclusive rational cup length of this manifold are both four. Thus the example disproves the unrestricted critical-number and rational cup-length forms of the Arnold conjecture.

1 Introduction

Arnold's fixed-point conjecture compares Hamiltonian dynamics with the critical-point theory of smooth functions. For a sufficiently short autonomous Hamiltonian flow on a closed symplectic manifold, its fixed points are exactly the critical points of its Hamiltonian. The conjecture asks whether an arbitrary Hamiltonian diffeomorphism must retain the minimum number of critical points forced by the topology of the manifold. Arnold's formulation explicitly includes the count of geometrically distinct points [1, Section 2]. We construct a smooth counterexample on the complex quadric threefold, which also violates the weaker rational cup-length bound.

The two degenerate bounds. For a closed smooth manifold M , write

$$\text{crit}(h) = \{q \in M : dh_q = 0\}, \quad \text{Crit}(M) = \min_{h \in C^\infty(M, \mathbb{R})} \# \text{crit}(h).$$

The minimum here permits degenerate critical points, whereas the *Morse number* minimizes only over Morse functions. Let $\text{cuplength}(M; \mathbb{Q})$ be one plus the maximum number of positive-degree rational cohomology classes with nonzero cup product. Thus our cup-length convention includes the unit. The Lusternik–Schnirelmann inequality gives

$$\text{cuplength}(M; \mathbb{Q}) \leq \text{Crit}(M);$$

Section 5 recalls the cup-product argument in a form that applies to arbitrary smooth functions.

A Hamiltonian diffeomorphism of a symplectic manifold (M, ω) is the time-one map of a smooth time-dependent Hamiltonian flow. We write $\text{Ham}(M, \omega)$ for these maps, use the convention $\iota_{X_{H_t}} \omega = dH_t$, and set $\text{Fix}(\phi) = \{q \in M : \phi(q) = q\}$. The *critical-number Arnold bound* asserts that

$$\# \text{Fix}(\phi) \geq \text{Crit}(M) \tag{1.1}$$

for every $\phi \in \text{Ham}(M, \omega)$ on every closed connected symplectic manifold. Its weaker *rational cup-length* form replaces $\text{Crit}(M)$ by $\text{cuplength}(M; \mathbb{Q})$. Neither formulation assumes nondegeneracy; see Golovko [7, Conjectures 1.4 and 1.6]. A fixed point q is nondegenerate when 1 is not an eigenvalue of $d\phi_q$.

Let

$$Q^3 = \{[z_0 : z_1 : z_2 : z_3 : z_4] \in \mathbb{CP}^4 : z_0^2 + z_1^2 + z_2^2 + z_3^2 + z_4^2 = 0\}, \quad (1.2)$$

with the restricted Fubini–Study symplectic form, normalized in Section 3.

Theorem 1.1. *There is a smooth Hamiltonian diffeomorphism ϕ of the closed connected symplectic six-manifold Q^3 such that*

$$\# \text{Fix}(\phi) = 3 < 4 = \text{Crit}(Q^3) = \text{cuplength}(Q^3; \mathbb{Q}). \quad (1.3)$$

At least one fixed point of ϕ is degenerate.

This disproves both unrestricted smooth bounds. The count includes all fixed points, so imposing a contractibility condition on their Hamiltonian trajectories cannot restore either inequality. The nondegenerate homological Arnold inequalities concern a different hypothesis and remain compatible with this example.

The count of three is sharp. Gong [8, Section 1.1.1] obtains at least n fixed points for every Hamiltonian diffeomorphism of the standard complex quadric of complex dimension $n \geq 2$. His normalization of the Fubini–Study form differs from ours by a positive scalar, which does not change the Hamiltonian group: rescaling the form and a generating Hamiltonian by the same scalar preserves its vector field. Thus our example attains his bound for $n = 3$.

Historical context and related work. The positive cases explain why the unrestricted comparison is natural. Conley and Zehnder proved that every Hamiltonian diffeomorphism of the standard symplectic torus T^{2n} has at least $2n + 1$ fixed points [3], and Fortune obtained the bound $n + 1$ on standard $\mathbb{CP}^n$ [5]. Hofer obtained cup-length bounds without nondegeneracy when $\pi_2(M) = 0$ [9, Theorem 4]. Rudyak and Oprea proved the full critical-number bound when both the symplectic class and the first Chern class vanish on $\pi_2(M)$ [17, Corollary 4.3]. The quadric has spheres of positive symplectic area and lies outside these latter hypotheses.

Floer’s pseudoholomorphic-curve methods established homological Morse inequalities for nondegenerate Hamiltonian fixed points, including the monotone setting [4]. The lower bound by the sum of rational Betti numbers was subsequently established for nondegenerate Hamiltonian diffeomorphisms on every closed symplectic manifold by Fukaya and Ono, Liu and Tian, and Ruan [6, 11, 16]. The distinction between these bounds and the degenerate formulations above is essential; see [7, Theorem 1.3 and Conjectures 1.4 and 1.6]. Ma [13, Theorem 1.5] asserted that the minimum number of fixed points over all Hamiltonian diffeomorphisms always equals $\text{Crit}(M)$. Theorem 1.1 contradicts that assertion.

Regularity also matters. Buhovsky, Humilière, and Seyfaddini constructed Hamiltonian *homeomorphisms* with a single fixed point on every closed connected symplectic manifold of real dimension at least four [2]. Our construction stays in the smooth Hamiltonian group. In a preprint posted September 27, 2026, Jiao [10, Theorem 1.1] reports a smooth Hamiltonian diffeomorphism of the equal-area product $S^2 \times S^2$ with two fixed points, using local saddle merging and mixed generating functions. The construction here instead uses a finite-order map on a six-dimensional quadric. None of these fixed-point results is an input to our construction.

Proof mechanism. The key is to transfer a critical-point count from a fixed submanifold to the fixed-point count of a nearby Hamiltonian map. Let A be a finite-order Hamiltonian map with fixed submanifold F , whose fixed tangent vectors are exactly those tangent to F . Extend a function f on

F to an A -invariant function K on the ambient manifold, and let B_s be its Hamiltonian flow. If $A^m = \text{id}$, invariance gives

$$(A \circ B_\varepsilon)^m = B_{m\varepsilon}.$$

For sufficiently small $\varepsilon > 0$, the right side has only stationary fixed points. Any fixed point of $A \circ B_\varepsilon$ is therefore a critical point of K fixed by A . Finite-group averaging makes ambient criticality on F equivalent to criticality of f , giving

$$\text{Fix}(A \circ B_\varepsilon) = \text{crit}(f).$$

This criticality step is the finite-group case of Palais's principle of symmetric criticality [15, Section 5]; the power identity excludes additional fixed points away from F .

For the quadric, take the Hamiltonian involution

$$A[z_0 : z_1 : z_2 : z_3 : z_4] = [z_0 : -z_1 : -z_2 : -z_3 : -z_4].$$

Its fixed locus is the quadric surface $Q^2 \cong S^2 \times S^2$. This product admits functions with only three critical points, as recorded by Mare [14, Proposition 3.2.5] through an extension of Takens's sphere-product construction [18]. We give an explicit formula and compute its entire critical set. The isolated degenerate critical point becomes a degenerate Hamiltonian fixed point. In contrast, symplectic volume forces every smooth function on the ambient Q^3 to have at least four critical points, and a Hamiltonian rotation realizes that minimum.

Section 2 proves the finite-order transfer, including the short-period estimate and invariant extension. Section 3 constructs the Hamiltonian involution and its tangent splitting. Section 4 gives the three-critical-point function. Section 5 proves the critical-number and rational cup-length equalities on Q^3 , and Section 6 assembles the construction and verifies degeneracy.

2 Perturbing a finite-order Hamiltonian map

A short Hamiltonian perturbation of a finite-order map can replace its fixed submanifold by the critical set of a function on that submanifold. The tangent-space hypothesis below is the usual *clean fixed-set* condition; we state it explicitly to identify the geometric input.

Proposition 2.1. *Let (M, ω) be a closed symplectic manifold, and let $A \in \text{Ham}(M, \omega)$ satisfy $A^m = \text{id}$ for some positive integer m . Suppose that $F = \text{Fix}(A)$ is a closed embedded submanifold and that*

$$T_q F = \ker(dA_q - \text{id}) \quad \text{for every } q \in F.$$

For every smooth function $f : F \rightarrow \mathbb{R}$, there is a smooth A -invariant extension $K : M \rightarrow \mathbb{R}$ such that, if B_s is its Hamiltonian flow, then for every sufficiently small $\varepsilon > 0$ the map

$$\phi_\varepsilon = A \circ B_\varepsilon$$

is Hamiltonian and satisfies

$$\text{Fix}(\phi_\varepsilon) = \text{crit}(f),$$

where the critical set on the right is viewed as a subset of M .

The proof combines finite-group averaging with the exclusion of short nonconstant periodic trajectories. We establish the latter fact first. Lipschitz control of a vector field gives a lower bound for its nonconstant periods, a classical principle for which Yorke proved the sharp Euclidean estimate [20]. The following elementary local argument suffices here.

Lemma 2.2. *Let X be a smooth vector field on a closed manifold M , and let Ψ_t be its flow. There is a number $\delta > 0$ such that*

$$\text{Fix}(\Psi_t) = \{q \in M : X(q) = 0\} \quad \text{for every } 0 < t < \delta.$$

Proof. The flow is defined for all time because M is compact. Suppose that there were points q_j and times $t_j > 0$ with

$$t_j \rightarrow 0, \quad \Psi_{t_j}(q_j) = q_j, \quad X(q_j) \neq 0.$$

After passing to a subsequence, assume that $q_j \rightarrow q$. Choose coordinates near q , and a Euclidean ball about its coordinate image whose closure lies inside the coordinate domain. By continuity of the flow, all sufficiently late trajectory segments $\{\Psi_s(q_j) : 0 \leq s \leq t_j\}$ lie in this ball. The coordinate vector field V is Lipschitz there with a constant $L \geq 0$, since its derivative is bounded on the closed ball.

Write $u_j(s)$ for the coordinate trajectory and set

$$C_j = \max_{0 \leq s \leq t_j} |\dot{u}_j(s)| > 0.$$

For any $s, r \in [0, t_j]$, convexity of the ball and the Lipschitz estimate give

$$\begin{aligned} |\dot{u}_j(s) - \dot{u}_j(r)| &= |V(u_j(s)) - V(u_j(r))| \\ &\leq L|u_j(s) - u_j(r)| \leq LC_j|s - r| \leq LC_j t_j. \end{aligned}$$

The closed trajectory has zero mean coordinate velocity: $\int_0^{t_j} \dot{u}_j(r) dr = 0$. At a time s_j of maximum speed, it follows that

$$C_j = \left| \frac{1}{t_j} \int_0^{t_j} (\dot{u}_j(s_j) - \dot{u}_j(r)) dr \right| \leq LC_j t_j.$$

Dividing by C_j contradicts $t_j \rightarrow 0$. Thus all fixed points of a sufficiently short positive-time map are zeros of X . Conversely, every zero of X is stationary and hence fixed by every Ψ_t . $\square$

Proof of Proposition 2.1. The first step is to extend f so that criticality along F is equivalent to ambient criticality at points of F . This is the finite-group case of symmetric criticality [15], which we verify directly. Extend f smoothly to a function $\tilde{f}$ on M : in submanifold charts, extend independently of the normal coordinates, and combine these local extensions using a partition of unity. Define

$$K = \frac{1}{m} \sum_{j=0}^{m-1} \tilde{f} \circ A^j.$$

Then $K \circ A = K$ and $K|_F = f$.

Fix $q \in F$, put $D = dA_q$, and consider

$$P_q = \frac{1}{m} \sum_{j=0}^{m-1} D^j.$$

Since $D^m = \text{id}$, we have $DP_q = P_q$. Moreover, $P_q v = v$ whenever $Dv = v$. Thus P_q is a projection onto $\ker(D - \text{id}) = T_q F$. Differentiating the averaged extension gives, for every $v \in T_q M$,

$$dK_q(v) = d\tilde{f}_q(P_q v) = df_q(P_q v).$$

Consequently,

$$\text{crit}(K) \cap F = \text{crit}(f). \quad (2.1)$$

Because A is symplectic and $K \circ A = K$, its differential carries the Hamiltonian vector field X_K to itself. Uniqueness of solutions therefore gives $A \circ B_s = B_s \circ A$ for every s . Apply Lemma 2.2 to X_K , and choose $0 < \varepsilon < \delta/m$. We then have

$$\phi_\varepsilon^m = B_{m\varepsilon}, \quad \text{Fix}(B_{m\varepsilon}) = \text{crit}(K).$$

If $\phi_\varepsilon(q) = q$, then q is fixed by $B_{m\varepsilon}$, so $dK_q = 0$. It follows that $B_\varepsilon(q) = q$, and hence $A(q) = q$. Conversely, every point of $\text{crit}(K) \cap F$ is fixed by ϕ_ε . Equation 2.1 proves the asserted equality of fixed and critical sets.

For completeness, the composition is generated by a smooth Hamiltonian on the unit time interval. Choose a Hamiltonian path A_t from id to A , generated by H_t , and a smooth nondecreasing function $a : [0, 1] \rightarrow [0, 1]$ equal to 0 near 0 and to 1 near 1. The time-dependent Hamiltonian

$$L_t(x) = \begin{cases} 2\varepsilon a'(2t)K(x), & 0 \leq t \leq \frac{1}{2}, \\ 2a'(2t-1)H_{a(2t-1)}(x), & \frac{1}{2} \leq t \leq 1 \end{cases}$$

is smooth, since both pieces vanish near the joining time. Its flow first follows $B_{\varepsilon a(2t)}$, and then $A_{a(2t-1)} \circ B_\varepsilon$, so its time-one map is ϕ_ε . $\square$

The proposition imposes no bound on the total number of critical points of K , and requires invariance under A , not under an isotopy from id to A . A critical point $q \notin F$ satisfies $\phi_\varepsilon(q) = A(q) \neq q$, so it contributes no fixed point.

3 A Hamiltonian involution of a quadric

We now construct the manifold and involution to which Proposition 2.1 will apply. Let $M \subset \mathbb{CP}^4$ be the quadric defined in (1.2).

Proposition 3.1. *The manifold M is closed, connected, and of real dimension six. For the restricted Fubini–Study form ω , normalized below, the map*

$$A[z_0 : z_1 : z_2 : z_3 : z_4] = [z_0 : -z_1 : -z_2 : -z_3 : -z_4]$$

is a Hamiltonian involution. Its fixed set is the smooth submanifold

$$F = M \cap \{z_0 = 0\} \cong S^2 \times S^2.$$

For every $q \in F$, the differential dA_q acts as $+\text{id}$ on $T_q F$ and as $-\text{id}$ on a complementary real two-dimensional subspace of $T_q M$.

Proof. In any standard affine chart of $\mathbb{CP}^4$, the equation defining M takes the form $1 + \sum_j w_j^2 = 0$. Its differential cannot vanish on its zero set: that would force all the w_j to be zero. Thus M is a smooth complex hypersurface, of complex dimension three. As a closed subset of projective space it is compact, and it has no boundary.

To see connectedness, write a nonzero isotropic vector as $z = x + iy$ with $x, y \in \mathbb{R}^5$. Its real and imaginary parts satisfy

$$|x| = |y| > 0, \quad x \cdot y = 0.$$

After rescaling, they form an ordered orthonormal pair. Every such pair extends to a positively oriented orthonormal basis, so these pairs form one orbit of the connected group $\mathrm{SO}(5)$. Their map $(x, y) \mapsto [x + iy]$ onto M proves that M is connected.

We construct the symplectic form with its normalization explicit. On the unit sphere $S^9 \subset \mathbb{C}^5$, put

$$\alpha = \sum_{j=0}^4 (x_j dy_j - y_j dx_j).$$

The form $d\alpha$ is invariant under scalar phase rotations. If $v \in T_z S^9$, then

$$d\alpha(iz, v) = -2 \operatorname{Re} \sum_{j=0}^4 \bar{z}_j v_j = 0.$$

Consequently it descends under the quotient map $p : S^9 \rightarrow \mathbb{CP}^4$ to a two-form σ , characterized by $p^* \sigma = d\alpha$. Local sections show that σ is smooth and closed. A projective tangent vector has a horizontal lift $v \in \mathbb{C}^5$ satisfying $\sum_j \bar{z}_j v_j = 0$; multiplication by i on these lifts induces the complex structure. Since

$$d\alpha(v, iv) = 2|v|^2,$$

σ is positive on every complex tangent line. Its restriction $\omega = \sigma|_M$ is therefore closed and nondegenerate.

We use a family of Hamiltonian rotations: equal speeds will give the involution, and unequal speeds will later give a function with four critical points on M . For $a, b, t \in \mathbb{R}$, let $\Phi_t^{a,b}$ keep z_0 fixed and act on $(z_1, z_2)^\top$ and $(z_3, z_4)^\top$ by R_{at} and R_{bt} , respectively, where

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

These real orthogonal, complex unitary transformations preserve the quadric and α . If $Z_{a,b}$ is their generating vector field on S^9 , Cartan's formula gives

$$\iota_{Z_{a,b}} d\alpha = -d(\alpha(Z_{a,b})).$$

The function $\alpha(Z_{a,b})$ is phase invariant, because the rotations commute with scalar multiplication. It thus descends to a smooth function $g_{a,b}$ on projective space. The displayed identity descends as well, so $G_{a,b} = -g_{a,b}|_M$ generates the projected rotations on M with our convention $\iota_{X_{G_{a,b}}} \omega = dG_{a,b}$. Explicitly,

$$G_{a,b}([z]) = \frac{2a \operatorname{Im}(\bar{z}_1 z_2) + 2b \operatorname{Im}(\bar{z}_3 z_4)}{\sum_{j=0}^4 |z_j|^2}. \quad (3.1)$$

In particular, $A = \Phi_1^{\pi,\pi}$ is Hamiltonian and preserves ω .

A projective line fixed by A lies in one of the two eigenspaces of $\operatorname{diag}(1, -1, -1, -1, -1)$. The positive eigenspace gives the single point $[1 : 0 : 0 : 0 : 0]$, which does not lie in M . The negative eigenspace gives precisely $F = M \cap \{z_0 = 0\}$. To identify F , use the invertible complex-linear map

$$(z_1, z_2, z_3, z_4) \mapsto \begin{pmatrix} z_1 + iz_2 & z_3 + iz_4 \\ -z_3 + iz_4 & z_1 - iz_2 \end{pmatrix}.$$

Its determinant is $z_1^2 + z_2^2 + z_3^2 + z_4^2$. It therefore identifies F with the projectivized nonzero rank-one matrices. The map

$$\mathbb{CP}^1 \times \mathbb{CP}^1 \longrightarrow F, \quad ([u], [v]) \mapsto [uv^\top]$$

is a diffeomorphism: a nonzero column and row recover the two projective factors, smoothly on the corresponding matrix charts. Finally, stereographic coordinates identify each $\mathbb{CP}^1$ with S^2 . Only this smooth identification is needed below.

For the tangent-space assertion, fix $q \in F$ and choose an affine chart $z_j = 1$ with $j > 0$. In its normalized coordinates, A sends w_0 to $-w_0$ and leaves all other coordinates fixed. The equation of M is

$$w_0^2 + 1 + \sum_{\substack{k>0 \\ k \neq j}} w_k^2 = 0.$$

At q , where $w_0 = 0$, its linearization places no constraint on the w_0 direction. The remaining differential is nonzero, so its kernel in the other coordinates is exactly $T_q F$. Thus

$$T_q M = T_q F \oplus \mathbb{C} \frac{\partial}{\partial w_0},$$

with the asserted $+1$ and -1 actions. This also verifies directly that F is a smooth complex surface. The positivity established above therefore makes $\omega|_F$ symplectic. $\square$

4 Three critical points on the fixed submanifold

We next construct a function on the fixed submanifold with three critical points. The comparison with functions on the ambient quadric will follow in Section 5. Three-critical-point functions on products of spheres belong to the classical study of critical-point minima and category by Takens [18]; the $S^2 \times S^2$ case is recorded by Mare [14, Proposition 3.2.5]. We use the explicit formula below and determine its entire critical set directly, without importing an existence theorem or assuming that its critical points are Morse.

Proposition 4.1. *Let $p = (0, 0, 1)$ and $e = (1, 0, 0)$ in $\mathbb{R}^3$. The smooth function*

$$f: S^2 \times S^2 \longrightarrow \mathbb{R}, \quad f(x, y) = e \cdot x + (x - p) \cdot y, \quad (4.1)$$

has exactly three critical points.

Proof. For a fixed $x \neq p$, the y -dependence is a height function on the second sphere, with critical points $y = \pm(x - p)/|x - p|$. At $x = p$ that height function is constant, but criticality of the full function still requires its derivative in the x directions to vanish. We therefore separate this exceptional fiber from the two height branches.

The constrained critical-point equations are

$$e + y = \lambda x, \quad x - p = \mu y, \quad |x| = |y| = 1, \quad (4.2)$$

for real multipliers λ, μ . If $x = p$, the first equation gives $y = -e + \lambda p$. Its unit norm forces $\lambda = 0$, yielding the unique point

$$q_0 = (p, -e), \quad f(q_0) = 0.$$

Suppose now that $x \neq p$ and set $t = |x - p| > 0$. The second equation in (4.2) gives $y = s(x - p)/t$ for a sign $s \in \{1, -1\}$. The first then becomes

$$te - sp = (\lambda t - s)x.$$

Since the left side is nonzero, $\lambda t - s \neq 0$ and x lies in the plane spanned by e and p . Write $x = (u, 0, w)$. The definition of t gives $w = 1 - t^2/2$. Taking the first equation in (4.2) against the tangent vector $(w, 0, -u)$ gives

$$w + \frac{su}{t} = 0, \quad \text{so} \quad u = -stw.$$

Consequently the unit-norm condition reduces to

$$0 = (1 - t^2/2)^2(1 + t^2) - 1 = \frac{t^4(t^2 - 3)}{4}. \quad (4.3)$$

Because $t > 0$, this forces $t = \sqrt{3}$, $w = -1/2$, and $u = s\sqrt{3}/2$. In particular, the other pole $x = -p$, for which $t = 2$, is excluded. The two resulting candidates are

$$q_s = \left(\left(\frac{s\sqrt{3}}{2}, 0, -\frac{1}{2} \right), \left(\frac{1}{2}, 0, -\frac{s\sqrt{3}}{2} \right) \right), \quad s \in \{1, -1\}. \quad (4.4)$$

Both satisfy (4.2) with $\lambda = \mu = s\sqrt{3}$, and

$$f(q_s) = s \frac{3\sqrt{3}}{2}.$$

The cases $x = p$ and $x \neq p$ exhaust the domain, proving the count. $\square$

Remark 4.2. The critical point q_0 is degenerate. Indeed, in local coordinates

$$x = (u, v, \sqrt{1 - u^2 - v^2}), \quad y = (-\sqrt{1 - a^2 - b^2}, a, b),$$

the function has expansion

$$f = va + O(\|(u, v, a, b)\|^3).$$

Its Hessian therefore has eigenvalues $1, -1, 0, 0$, so its rank is two. The complete critical-point calculation proves that this degenerate point is isolated. This is why a count restricted to Morse functions would not capture the example.

5 The critical number and cup length of the quadric

We now show that four is both the critical number and the rational cup length of the ambient quadric. The lower bound is the classical cup-product obstruction from Lusternik–Schnirelmann theory [12]; see Weber [19, Theorems 3.5 and 3.6] for a modern treatment valid without a Morse assumption. We give the differential-form proof, then obtain the matching upper bound from the unequal-speed rotations of Section 3.

Lemma 5.1. *Let M be a closed smooth manifold and $r \geq 1$ an integer. Suppose there are classes $a_j \in H^{d_j}(M; \mathbb{R})$, with $d_j > 0$ for $1 \leq j \leq r$, such that $a_1 \smile \cdots \smile a_r \neq 0$. Then every smooth function on M has at least $r + 1$ critical points.*

Proof. Suppose that $h : M \rightarrow \mathbb{R}$ has at most r critical points. List its distinct critical values as $c_1 < \cdots < c_k$, where $1 \leq k \leq r$; the absolute minimum and maximum are among them. For a fixed Riemannian metric let g_t be the negative-gradient flow of h , which is complete by compactness, and put $P_b = \{h \leq b\}$. We first cover M by k open sets, each diffeomorphic to a finite disjoint union of

coordinate balls. Every closed positive-degree form will be exact on each of these sets; the cover will therefore force every product of k positive-degree cohomology classes to vanish.

Around the critical points at c_i , choose an open set U_i that is a union of disjoint coordinate balls, and a smaller neighborhood V_i with $\bar{V}_i \subset U_i$. A bound on the speed of the flow and the positive distance between $M \setminus U_i$ and $\bar{V}_i$ give $T_i > 0$ such that trajectories starting outside U_i stay outside V_i for $0 \leq t \leq T_i$.

Choose a closed level band $\{|h - c_i| \leq \delta_i\}$ narrow enough that it contains no critical points outside V_i . On the part outside V_i , compactness gives a positive lower bound b_i for $|\nabla h|^2$; if that set is empty, any positive b_i will do. Choose $0 < \epsilon_i < \delta_i$ so that the intervals $[c_i - \epsilon_i, c_i + \epsilon_i]$ are separated and $2\epsilon_i < T_i b_i$. Along the negative-gradient flow,

$$\frac{d}{dt}h(g_t(q)) = -|\nabla h(g_t(q))|^2.$$

A point of $P_{c_i+\epsilon_i} \setminus U_i$ either already lies below $c_i - \epsilon_i$ or decreases at rate at least b_i until it reaches that level. It follows that

$$g_{T_i}(P_{c_i+\epsilon_i} \setminus U_i) \subset P_{c_i-\epsilon_i}. \quad (5.1)$$

For $i > 1$, a further fixed flow time $S_i > 0$ carries $P_{c_i-\epsilon_i}$ into $P_{c_{i-1}+\epsilon_{i-1}}$: the compact band between these levels contains no critical points and has a positive gradient-norm lower bound.

Since c_1 is the absolute minimum, $P_{c_1-\epsilon_1}$ is empty, and (5.1) shows that $P_{c_1+\epsilon_1} \subset U_1$. Inductively, cover $P_{c_i+\epsilon_i}$ by U_i and the preimages under $g_{T_i+S_i}$ of the $i-1$ sets covering the preceding upper sublevel. Equation (5.1) proves that these sets cover. Each g_t is a diffeomorphism of M , so every member remains diffeomorphic to a finite disjoint union of coordinate balls. The last upper sublevel is all of M , since c_k is the absolute maximum.

Write the resulting cover as $O_1, \dots, O_k$. Choose closed forms α_j representing a_j for $1 \leq j \leq k$. Since these forms have positive degree, the Poincaré lemma on each ball gives $\alpha_j = d\lambda_j$ on O_j . Choose a smooth partition of unity $\psi_1, \dots, \psi_k$ subordinate to the cover. Let $\chi : [0, 1] \rightarrow [0, 1]$ be smooth, zero on $[0, 1/(4k)]$ and one on $[1/(2k), 1]$, and put $\rho_j = \chi(\psi_j)$. At every point some $\psi_j \geq 1/k$, so some ρ_j is identically one nearby. Each $\rho_j \lambda_j$ extends smoothly by zero outside O_j .

The closed forms

$$\alpha'_j = \alpha_j - d(\rho_j \lambda_j)$$

represent the same classes as α_j . Near every point, at least one of these forms vanishes, and hence $\alpha'_1 \wedge \dots \wedge \alpha'_k = 0$. Thus $a_1 \smile \dots \smile a_k = 0$, contradicting the nonvanishing of $a_1 \smile \dots \smile a_r$ because $k \leq r$. $\square$

Extension of coefficients $H^*(M; \mathbb{Q}) \rightarrow H^*(M; \mathbb{R})$ is injective for a closed manifold and respects cup products. The lemma therefore also gives

$$\text{cuplength}(M; \mathbb{Q}) \leq \text{Crit}(M), \quad (5.2)$$

with the unit-inclusive convention of Section 1. It also recovers the symplectic-volume obstruction used in our example.

Corollary 5.2. *Every smooth function on a closed connected symplectic manifold of real dimension $2n$, with $n \geq 1$, has at least $n + 1$ critical points.*

Proof. The class $[\omega]^n$ is nonzero, since $\int_M \omega^n > 0$ in the symplectic orientation. Apply Lemma 5.1 with each $a_j = [\omega]$. $\square$

Corollary 5.3. *For the quadric threefold in (1.2),*

$$\text{Crit}(Q^3) = \text{cuplength}(Q^3; \mathbb{Q}) = 4.$$

Proof. Let $h = c_1(\mathcal{O}_{Q^3}(1)) \in H^2(Q^3; \mathbb{Z})$ be the hyperplane class. Since Q^3 has degree two in $\mathbb{CP}^4$,

$$h^3 = 2[\text{pt}],$$

where $[\text{pt}]$ evaluates to one on the fundamental class. This product remains nonzero over $\mathbb{Q}$, giving $\text{cuplength}(Q^3; \mathbb{Q}) \geq 4$.

For the matching upper bound, take the Hamiltonian $G_{1,2}$ of (3.1). Its critical points are precisely the zeros of its Hamiltonian vector field. That field is induced on Q^3 by the complex-linear generator

$$T = \text{diag}(0, J, 2J), \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The induced projective vector field vanishes at $[z]$ exactly when $Tz \in \mathbb{C}z$. The eigenvalues $0, \pm i, \pm 2i$ of T are distinct, so its projective zeros are its five eigenlines. The zero eigenline $[1 : 0 : 0 : 0 : 0]$ misses Q^3 , whereas the other four lines

$$[0 : 1 : \pm i : 0 : 0], \quad [0 : 0 : 0 : 1 : \pm i]$$

all satisfy the quadric equation. Thus $G_{1,2}$ has exactly four critical points on Q^3 . Together with (5.2), this gives

$$4 \leq \text{cuplength}(Q^3; \mathbb{Q}) \leq \text{Crit}(Q^3) \leq 4.$$

□

6 The Hamiltonian counterexample

We now apply the finite-order reduction to the function on the fixed surface and compare the resulting fixed-point count with the ambient critical number and cup length.

Proof of Theorem 1.1. Take the symplectic quadric (M, ω) and Hamiltonian involution A from Proposition 3.1. Its fixed submanifold F is diffeomorphic to $S^2 \times S^2$. Transport the function of Proposition 4.1 to F ; it has exactly three critical points.

Proposition 2.1 provides a smooth A -invariant extension $K: M \rightarrow \mathbb{R}$ with $K|_F = f$. Let B_s denote its Hamiltonian flow. For all sufficiently small $\varepsilon > 0$, the map

$$\phi = A \circ B_\varepsilon$$

is a Hamiltonian diffeomorphism and satisfies $\text{Fix}(\phi) = \text{crit}(f)$, with the critical points viewed as points of F . The exact comparison in Corollary 5.3 now gives

$$\# \text{Fix}(\phi) = 3 < 4 = \text{Crit}(M) = \text{cuplength}(M; \mathbb{Q}).$$

It remains to verify the asserted degeneracy. Write $\omega_F = \omega|_F$, which is symplectic, and let X_f be the Hamiltonian vector field of f on (F, ω_F) . Invariance of K makes X_K tangent to F , and restricting its defining equation to TF gives $X_K|_F = X_f$. Indeed, at $q \in F$ invariance gives $dA_q X_K(q) = X_K(q)$, and the fixed tangent space is $T_q F$. Thus B_s preserves F , and its restriction is the Hamiltonian flow of f on F . At the critical point q_0 of Remark 4.2, differentiating $\iota_{X_f} \omega_F = df$ gives

$$\omega_F(DX_f(q_0)v, w) = \text{Hess}_{q_0} f(v, w).$$

Thus any nonzero vector v in the Hessian kernel satisfies $DX_f(q_0)v = 0$. Since $A|_F = \text{id}$ and q_0 is an equilibrium,

$$d\phi_{q_0}v = \exp(\varepsilon DX_f(q_0))v = v.$$

The fixed point q_0 therefore has eigenvalue 1 in its linearization. This argument takes place entirely in the invariant subspace $T_{q_0}F$; no assumption on the normal Hessian of the extension K is needed. $\square$

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